Cliffe Owino Ibande

Software Engineer · Backend & AI Platforms

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Software engineer building retrieval-augmented AI platforms and full-stack products on the M-Pesa payments network. I have shipped four internal platforms for the engineering and support teams behind M-Pesa across five African markets, and four live products of my own, one of them moving real money over M-Pesa. Strongest in Python and FastAPI backends, and in retrieval that has to stay grounded in its sources. Looking for product engineering work, backend or full-stack, remote.

EXPERIENCE

Software Engineer, Core Payment Services — Prodapt Ltd

Sep 2025 – Present

M-Pesa Africa across Tanzania, Ghana, Mozambique, Lesotho and the DR Congo, under Prodapt's partnership with Vodacom. Platforms serving millions of users at 1M+ transactions a day.

PLATFORMS BUILT

- **FintechHub** — learning platform and knowledge assistant for a mobile-money platform upgrade; working MVP. Ingests vendor help archives, PDF, Word and PowerPoint into Milvus and a MySQL knowledge graph, inside a sandbox. Two reviewers sign off each module before it publishes, and learners earn reproducible PDF certificates a third party can verify. The assistant streams cited answers over SSE, reranks them and checks them for faithfulness; a release ships only if it clears recall@12 against a golden question set. **1,100+ backend tests at ~91% coverage, 670+ frontend.**
- **RCA Pulsar** — retrieval-augmented root cause analysis for M-Pesa problem management; team project. Fuses dense vectors with BM25 so exact error codes, ticket IDs and hostnames still rank. Four grounding checks run in order, and the system declines rather than invent a cause it cannot ground. Approved analyses move through a maker-checker state machine where no one approves a document they wrote, and approval re-indexes it so it informs later answers. **1,570 backend tests.**
- **AI LMS** — turns a support team's SOPs into courses and assessments; team project. Generates courses with LangChain and checks that each question can be answered from its own lesson. An hourly job confirms a changed SOP across several runs, then regenerates the lesson into staging for the owner to review. Assessments shuffle from a fixed seed, so I can rebuild any paper later, and a reversible PII vault keeps real hostnames, IPs and phone numbers away from the model. **1,600+ backend tests, 185+ frontend.**
- **OpsHub** — the hub running the CPS support team's on-call, rotas, incidents and planned changes. Drafts the daily rota from fairness rules for a team lead to review, approve and publish. The incident assistant combines BM25 and vector search over SOPs and ticket history to return cited triage steps and a draft worklog, masking phone numbers, emails, IPs and hostnames before anything leaves for the model. **500+ backend tests, permission coverage on every route.**

PLATFORM OPERATIONS

- Operate large-scale Linux production servers carrying M-Pesa core payment infrastructure at over 1M daily transactions
- Monitor, tune and optimise queries on high-volume transactional databases, **cutting latency 15%**
- Automate recurring operational work in Python and Bash, **cutting manual errors 30%**
- Create, renew and deploy the certificates securing service-to-service traffic, **sustaining 99.9% uptime**
- Provision VMs across distributed infrastructure on VMware and Kubernetes, and watch system health and SLA compliance in Grafana; handle incidents, restore service and run root-cause analysis with other teams
- Contribute to TZ Fintech 2.0, a cloud platform upgrade adding AI-driven fraud detection and transaction capacity

Backend Web Developer — Techsavannah

Sep 2024 – Jul 2025

Bespoke software and IT consulting for clients across Kenya.

- Shipped microservice architectures, Dockerised applications and Kubernetes deployments **across 10+ client projects**
- Built high-availability systems with API gateways, reverse proxies and load balancing on NGINX, **at 99% availability**
- Implemented Linux server automation, scheduled jobs, monitoring dashboards and deployment pipelines in Jenkins and Terraform
- Modernised legacy systems and optimised business processes alongside cross-functional teams

Data Analyst, Industrial Attachment — Huawei Technologies (Kenya)

Jun – Aug 2023

- Designed data models and maintained data cubes for network performance analytics
- Built dashboards in Python and Excel, and produced network monitoring reports flagging performance anomalies
- Analysed network performance trends in SPSS, Python and Excel; worked with cloud architecture, virtualisation and distributed systems

SELECTED PROJECTS

PichaKonnect — photography marketplace, live — Mar 2024 – Present

pichakonnect.live · React, Express, Prisma, PostgreSQL, Socket.IO, M-Pesa Daraja, Cloudinary

- Two-sided marketplace for Kenya: clients book photographers, photographers manage availability and deliver paid galleries, and **real money moves between them over M-Pesa**
- Built M-Pesa Daraja STK push with a verified callback and a duplicate-prompt guard, over a ledger carrying payouts, refunds and disputes
- Signed Cloudinary assets behind payment-gated galleries, Socket.IO chat, Google and Facebook sign-in, audit log on every admin action
- 81 backend tests and 155 commits; I wrote up three rounds of self-audit in the repository

Shorts Pipeline — video tooling, local models — Oct 2026 – Present

Python, faster-whisper, MediaPipe, OpenCV, FFmpeg, Pytest

- Command-line tool turning long-form interviews into captioned 9:16 video — transcription, clip selection, speaker-tracked reframing, captions and a voiced intro
- Runs on local models end to end: faster-whisper for transcription, MediaPipe face detection and landmarks to keep the speaker in a recropped frame, scene detection for cut points
- Every stage checkpoints its results, because a two-hour source takes three and a half hours and runs get interrupted; ~3,400 lines of Python, 13 test files

Kicksawa — e-commerce, live — Aug 2024 – Present

kicksawa.store · React, Redux Toolkit, Express, MongoDB, Firebase, Tailwind

- Sneaker and streetwear store built as three packages — customer storefront, admin dashboard and REST API — still shipping after two years at 73 commits
- Firebase authentication with JWT sessions; I hardened the backend with helmet and rate limiting, and built an admin dashboard reporting on orders and revenue

Royal Watches — e-commerce, live — Jun 2024 – Jun 2026

royalwatcheske.onrender.com · React, Redux Toolkit, Express, MongoDB, Stripe, Paystack

- Luxury watch storefront with a separate admin package, Stripe and Paystack payments, reviews and transactional email
- Audited it end to end in 2026 and **fixed 20 backend issues**, documenting the findings, API reference and remaining work in the repository

TECHNICAL SKILLS

LANGUAGES	Python, TypeScript, JavaScript, Java, SQL, Bash
BACKEND	FastAPI, SQLAlchemy 2, Alembic, Celery, Express, Node.js, Spring Boot, Flask, Django, REST, gRPC
FRONTEND	React, TypeScript, Next.js, Vite, TanStack Query, Zustand, Redux Toolkit, Tailwind, Radix
RETRIEVAL	Milvus, vector search, BM25 hybrid retrieval, LangChain, Azure OpenAI, Hugging Face, embeddings, RAG evaluation, PyTorch, scikit-learn
DATA	MySQL, PostgreSQL, MongoDB, Redis, Prisma, Supabase, Pandas, NumPy
INFRASTRUCTURE	Linux, Docker, Kubernetes, NGINX, Terraform, Ansible, Jenkins, GitHub Actions, VMware, Huawei Cloud, AWS, Azure, GCP
TESTING	Pytest, Vitest, Jest, Playwright, Schemathesis · Grafana, Prometheus, Loki, ELK Stack
SECURITY	Keycloak (OIDC), OAuth 2.0, JWT, role-based access control, PII masking and tokenisation

EDUCATION

BSc Software Engineering — Kisii University, Kenya — 2020 – Dec 2024

CERTIFICATIONS

Generative AI and LLMs: Architecture and Data Preparation — IBM / Coursera, 2026 (coursera.org/verify/TUQFDODE6BEQ)
Docker Fundamentals, 2025 · Postman API Development, 2025 · SPSS for Data Analysis, 2024 · Huawei Cloud Fundamentals (HCIA), 2023